Reconstruction fix for EL046, EL048 and PAT_3975

Where their electrodes were drawn on fsaverage before the fix, and where they are now. 12 November 2026

What was wrong

For these three patients the surf folder held a pial surface and a sphere.reg from two different reconstructions.
250 maps a contact to fsaverage through the vertex number it shares between those two files, so every contact that is snapped to the pial surface was sent to an unrelated place in the same hemisphere. Contacts routed through the affine (white-matter contacts) were never affected.

What changed

patient	file replaced in the surf folder	contacts	moved	unchanged	median move	largest	neighbours before	after	in cohort	moved
EL046	pial of the 18 Aug run (matches its sphere.reg)	178	166	12	59 mm	99 mm	40 mm	1.2 mm	47	47
EL048	pial of the 18 Aug run (matches its sphere.reg)	78	67	11	46 mm	112 mm	48 mm	4.3 mm	35	35
PAT_3975	its own sphere.reg (it held PAT_3965's)	212	164	48	46 mm	107 mm	37 mm	3.0 mm	110	76

All three: 468 contacts, 397 moved. In the clustering cohort (1587 electrodes, 29 patients): 192 electrodes of these patients, 158 of them moved (10 % of the cohort).

“neighbours” = median distance on fsaverage between consecutive contacts of one shaft, over the moved contacts (the other 40 patients: 2 to 10 mm).

What did not change

Which cluster an electrode belongs to: the clustering uses the ERSP, not the position.
The other 40 patients in the table: all 6118 contacts are at identical coordinates.
The patients' own space (tkrRAS coordinates): only the step to fsaverage was wrong.

Electrodes drawn at a wrong position, per cluster of the main solution: convex NMF, 15 bands, K = 7

	type	electrodes	patients	EL046	EL048	PAT_3975	moved	of the cluster	share of the cluster (axis to 30 %)
c0	beta decrease	316	21	3	0	15	18	6 %	
c1	audio-stimulus increase	135	21	1	0	2	3	2 %	
c2	broadband decrease	164	11	1	5	17	23	14 %	
c3	sentence-stimulus increase	194	25	1	8	24	33	17 %	
c4	low-frequency increase	349	29	15	10	12	37	11 %	
c5	response broadband increase	206	18	2	8	1	11	5 %	
c6	response low-freq. decrease	223	23	24	4	5	33	15 %	



Total: 158 of 1587 electrodes.

The same count for convex NMF, 15 bands each z-scored: c0 1/126 c1 23/217 c2 62/112 c3 18/204 c4 23/322 c5 20/299 c6 11/307

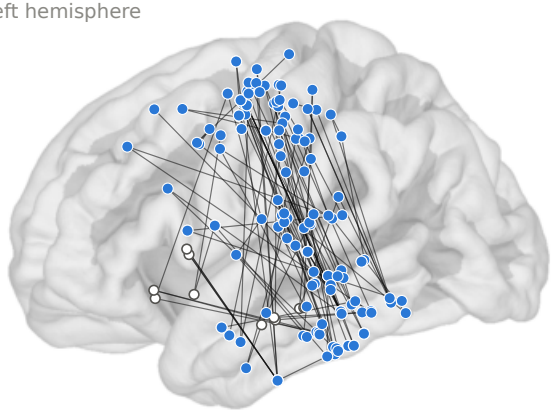
Pages 2 to 4: each patient. Pages 5 to 11: each cluster of the main solution. Pages 12 to 18: each cluster of the z-scored solution. Page 19: every run.

EL046 every contact in the table, joined along its shaft

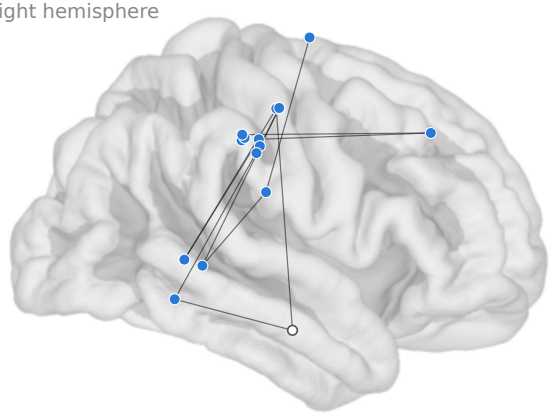
178 contacts · 166 snapped to the pial surface, all moved (median 59 mm, largest 99 mm) · 12 through the affine, unchanged

BEFORE positions in the table of 12 Oct: a shaft's contacts are scattered over the hemisphere

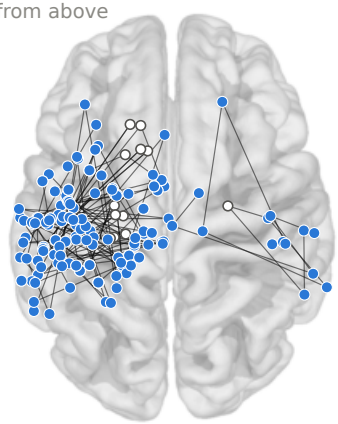
left hemisphere



right hemisphere

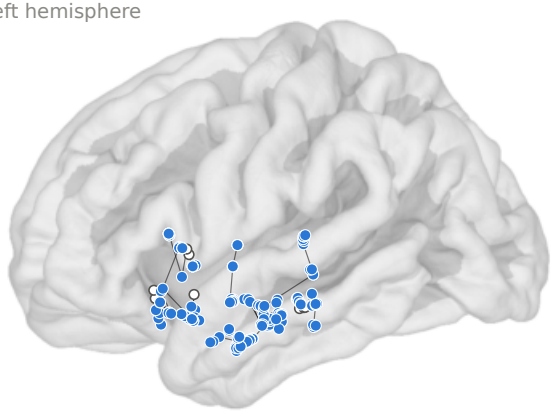


from above

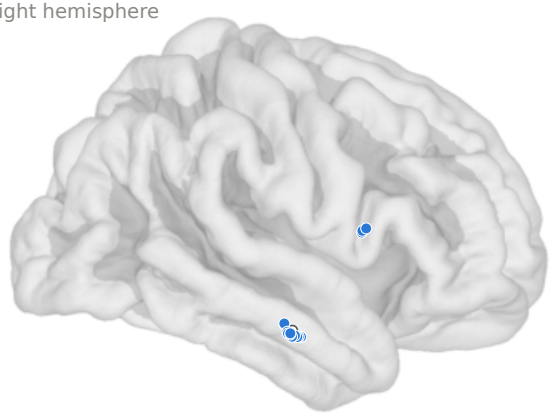


AFTER positions in the table of 12 Nov: the contacts of a shaft lie next to each other

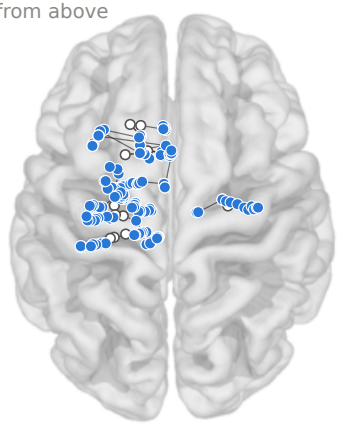
left hemisphere



right hemisphere



from above



Numbers

contacts in the table	178
moved by the fix	166
unchanged (affine route)	12
median move	59 mm
largest move	99 mm
neighbours on a shaft, before	40 mm
neighbours on a shaft, after	1.2 mm
electrodes in the cohort	47
of which moved	47

In the main solution

cluster	electrodes	moved
c0	3	3
c1	1	1
c2	1	1
c3	1	1
c4	15	15
c5	2	2
c6	24	24

● contact snapped to the pial surface (moved by the fix) ○ contact through the affine (unchanged) —— next contact on the shaft

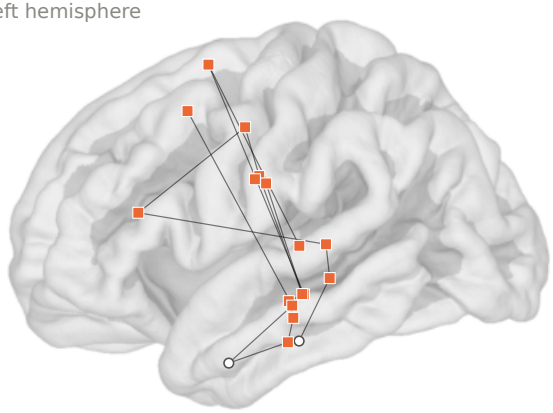
before = fsaverage table of 12 Oct 2026 (commit 766cc9571) · after = table written by 250 on 12 Nov 2026, 18:09 · lateral views show the electrodes of that hemisphere

EL048 every contact in the table, joined along its shaft

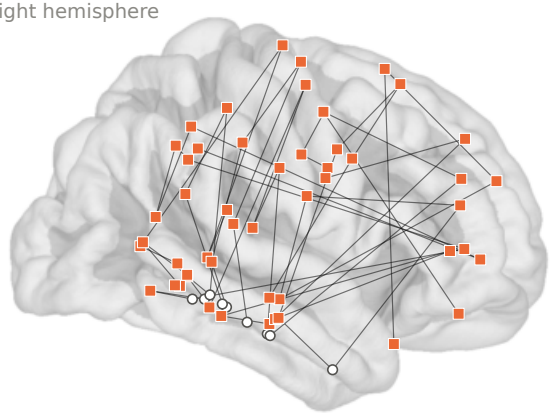
78 contacts · 67 snapped to the pial surface, all moved (median 46 mm, largest 112 mm) · 11 through the affine, unchanged

BEFORE positions in the table of 12 Oct: a shaft's contacts are scattered over the hemisphere

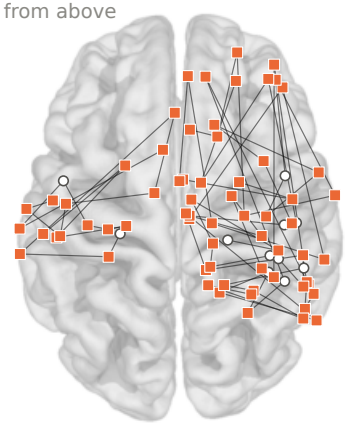
left hemisphere



right hemisphere

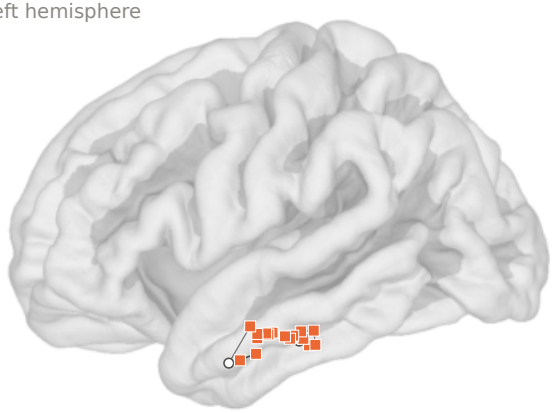


from above

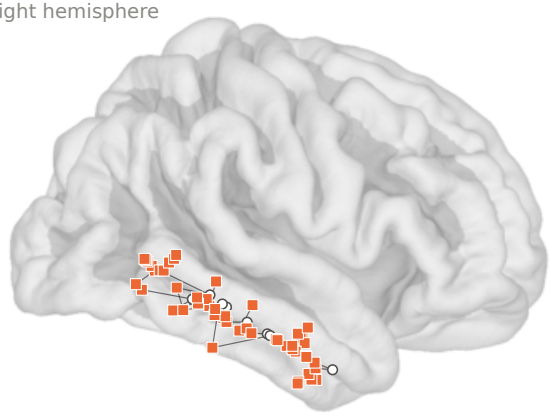


AFTER positions in the table of 12 Nov: the contacts of a shaft lie next to each other

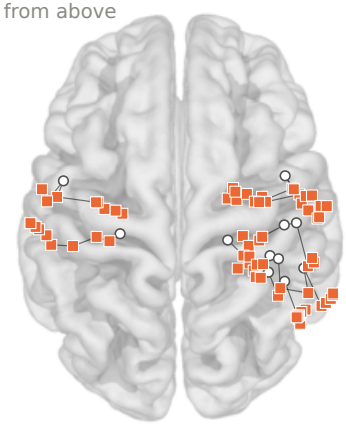
left hemisphere



right hemisphere



from above



■ contact snapped to the pial surface (moved by the fix) ○ contact through the affine (unchanged) —— next contact on the shaft

Numbers

contacts in the table	78
moved by the fix	67
unchanged (affine route)	11
median move	46 mm
largest move	112 mm
neighbours on a shaft, before	48 mm
neighbours on a shaft, after	4.3 mm
electrodes in the cohort	35
of which moved	35

In the main solution

cluster	electrodes	moved
c0	0	0
c1	0	0
c2	5	5
c3	8	8
c4	10	10
c5	8	8
c6	4	4

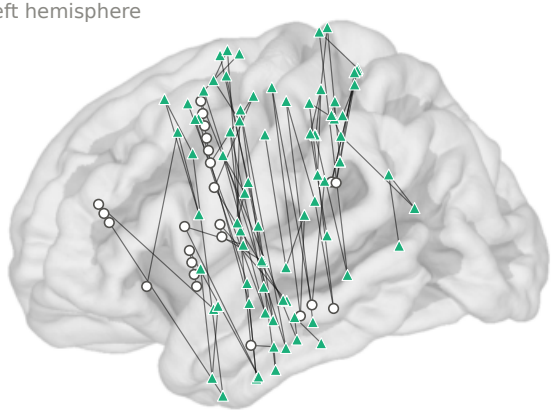
before = fsaverage table of 12 Oct 2026 (commit 766cc9571) · after = table written by 250 on 12 Nov 2026, 18:09 · lateral views show the electrodes of that hemisphere

PAT_3975 every contact in the table, joined along its shaft

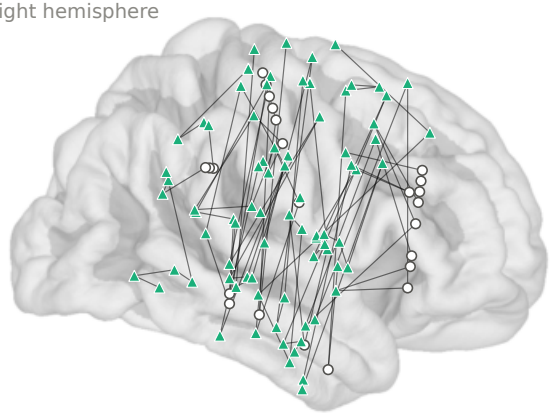
212 contacts · 164 snapped to the pial surface, all moved (median 46 mm, largest 107 mm) · 48 through the affine, unchanged

BEFORE positions in the table of 12 Oct: a shaft's contacts are scattered over the hemisphere

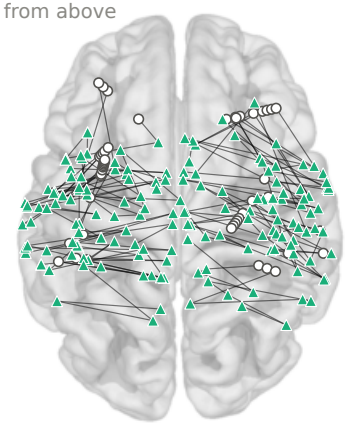
left hemisphere



right hemisphere

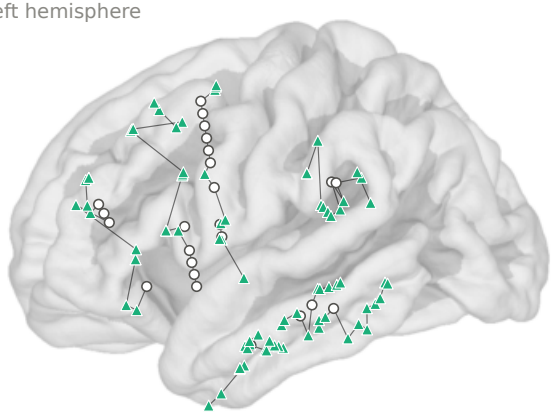


from above

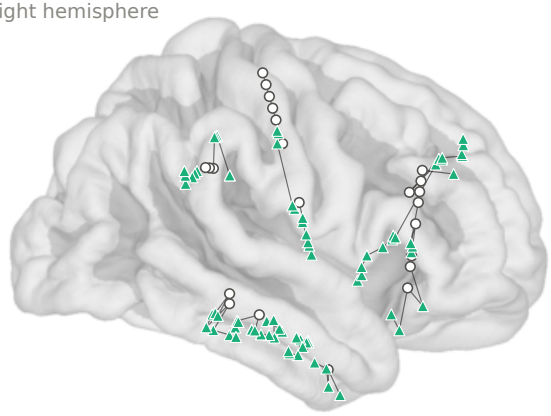


AFTER positions in the table of 12 Nov: the contacts of a shaft lie next to each other

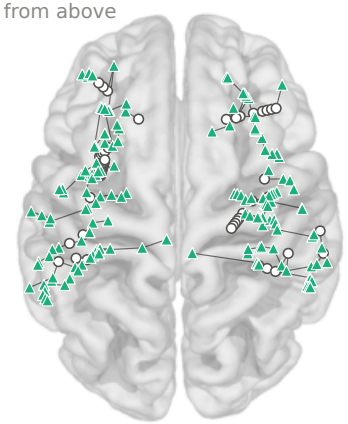
left hemisphere



right hemisphere



from above



▲ contact snapped to the pial surface (moved by the fix) ○ contact through the affine (unchanged) —— next contact on the shaft

Numbers

contacts in the table	212
moved by the fix	164
unchanged (affine route)	48
median move	46 mm
largest move	107 mm
neighbours on a shaft, before	37 mm
neighbours on a shaft, after	3.0 mm
electrodes in the cohort	110
of which moved	76

In the main solution

cluster	electrodes	moved
c0	29	15
c1	4	2
c2	20	17
c3	35	24
c4	14	12
c5	1	1
c6	7	5

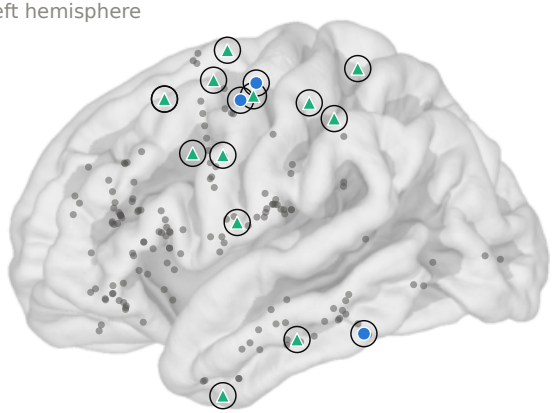
c0 beta decrease

main solution: convex NMF, 15 bands, K = 7

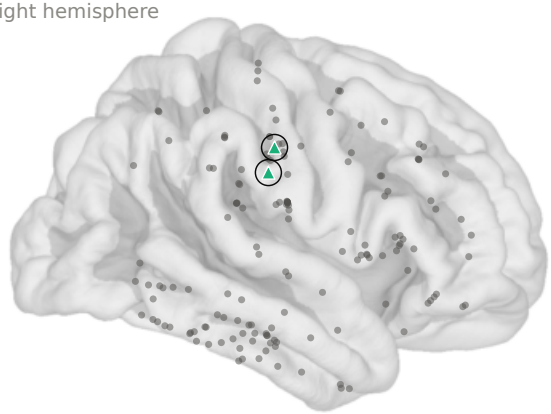
316 electrodes from 21 patients · 32 belong to the three patients · 18 were drawn at a wrong position (6 % of the cluster), median move 53 mm

BEFORE the cluster as it was drawn until now

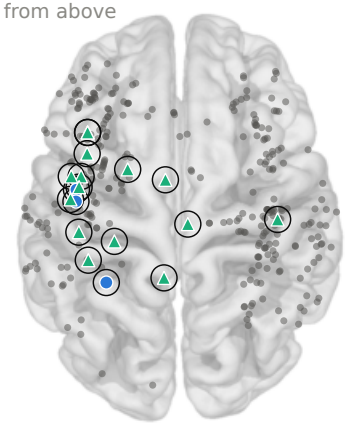
left hemisphere



right hemisphere

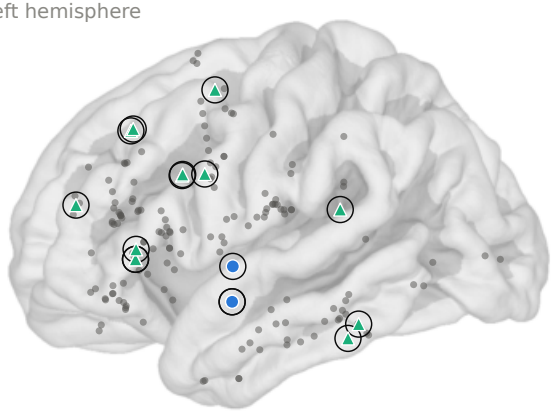


from above

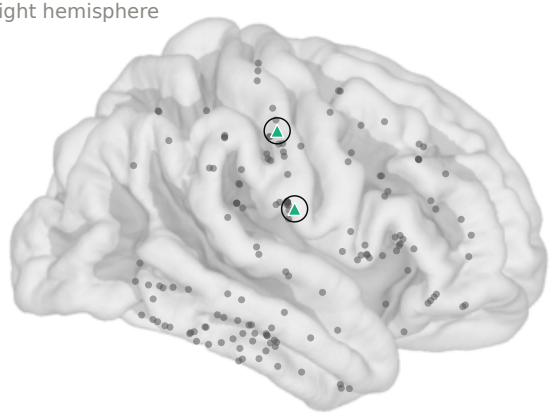


AFTER the cluster with the corrected positions

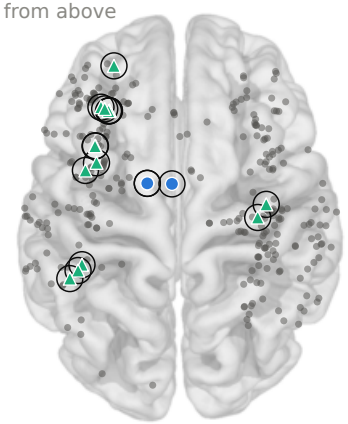
left hemisphere



right hemisphere



from above



- other electrodes of the cluster
- EL046
- EL048
- PAT_3975
- circled = position changed by the fix

Circled electrodes

patient	in the cluster	moved
EL046	3	3
EL048	0	0
PAT_3975	29	15

Where they are drawn

Desikan label of the nearest pial vertex

region	before	after
precentral	6	1
caudal middle frontal	3	4
medial wall (no label)	–	3
rostral middle frontal	–	3
inferior temporal	2	1
lateral orbitofrontal	–	2
postcentral	2	–
superior frontal	2	–
fusiform	1	1
5 more regions	2	3

The whole cluster, by lobe

electrodes

lobe	before	after
frontal	131	130
temporal	80	79
parietal	42	41
occipital	4	4
insula	44	45
cingulate	7	6
medial wall (no label)	8	11

c1 audio-stimulus increase

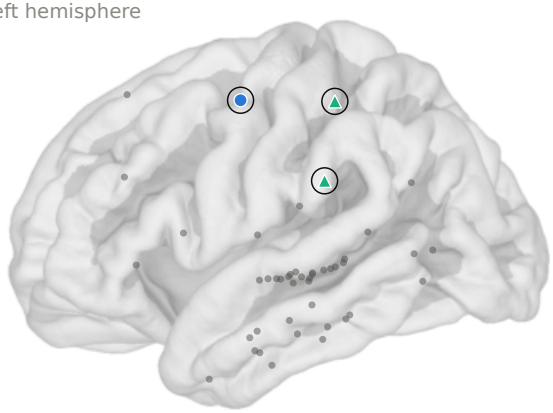
main solution: convex NMF, 15 bands, K = 7

135 electrodes from 21 patients · 5 belong to the three patients · 3 were drawn at a wrong position (2 % of the cluster), median move 67 mm

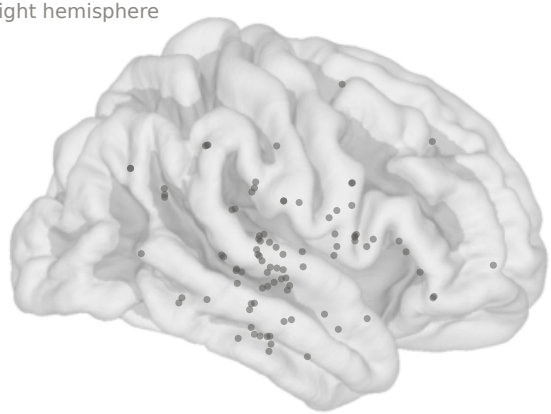
BEFORE

the cluster as it was drawn until now

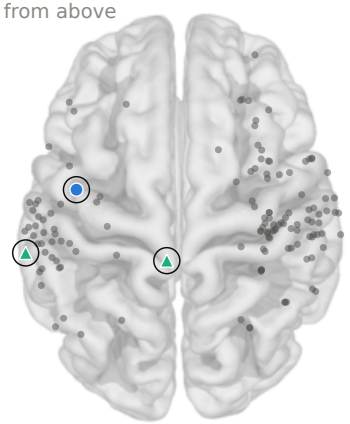
left hemisphere



right hemisphere



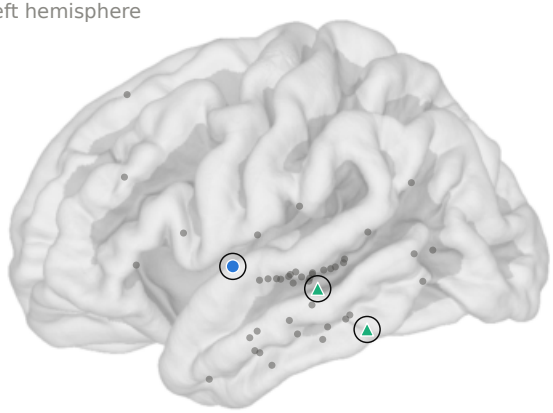
from above



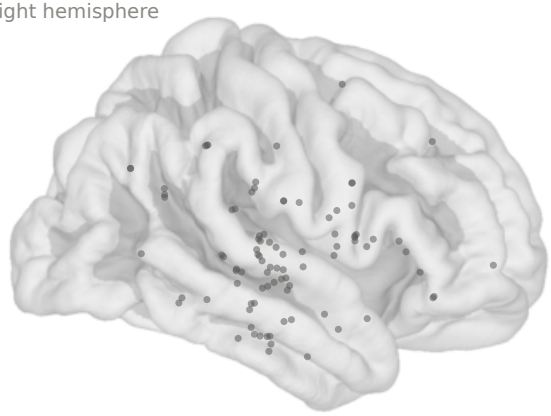
AFTER

the cluster with the corrected positions

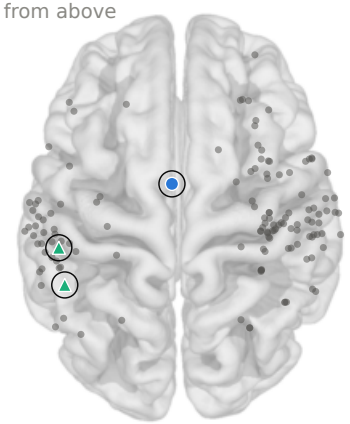
left hemisphere



right hemisphere



from above



- other electrodes of the cluster
- EL046
- EL048
- PAT_3975
- circled = position changed by the fix

Circled electrodes

patient	in the cluster	moved
EL046	1	1
EL048	0	0
PAT_3975	4	2

Where they are drawn

Desikan label of the nearest pial vertex

region	before	after
inferior temporal	–	1
medial wall (no label)	–	1
middle temporal	–	1
paracentral	1	–
precentral	1	–
supramarginal	1	–

The whole cluster, by lobe

electrodes

lobe	before	after
frontal	23	21
temporal	67	69
parietal	17	16
occipital	3	3
insula	18	18
medial wall (no label)	7	8

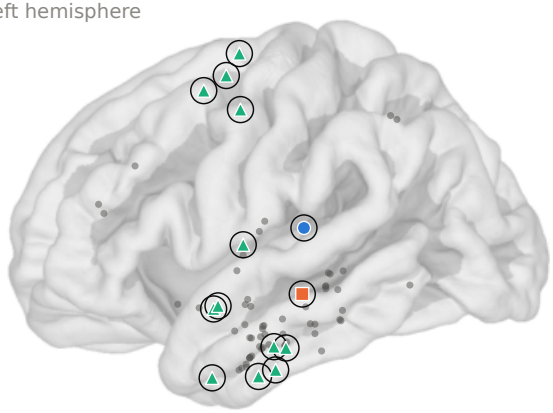
c2 broadband decrease

main solution: convex NMF, 15 bands, K = 7

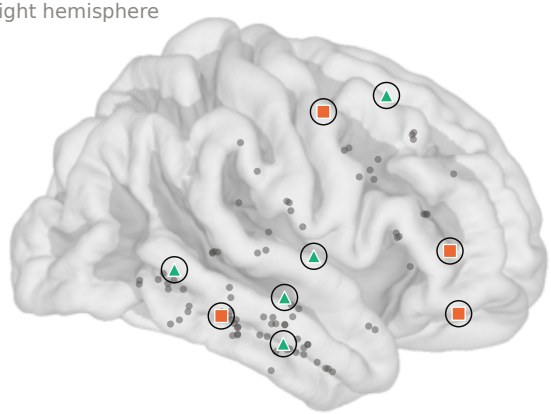
164 electrodes from 11 patients · 26 belong to the three patients · 23 were drawn at a wrong position (14 % of the cluster), median move 52 mm

BEFORE the cluster as it was drawn until now

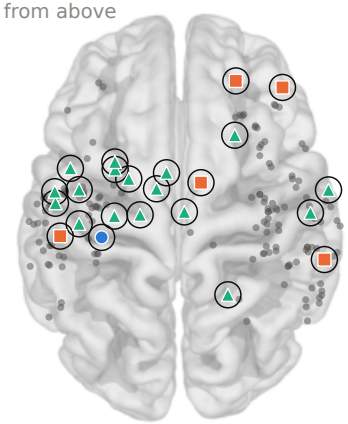
left hemisphere



right hemisphere

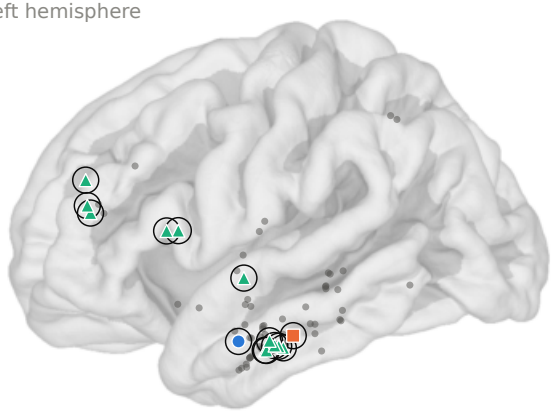


from above

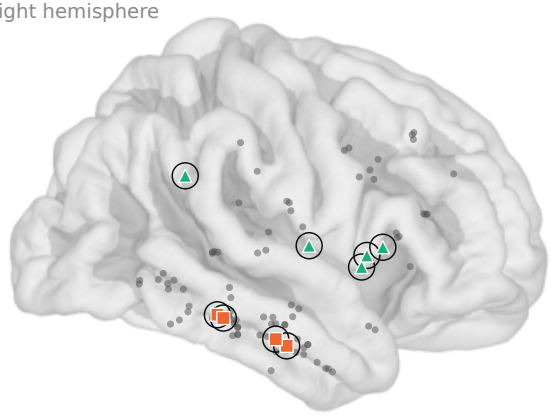


AFTER the cluster with the corrected positions

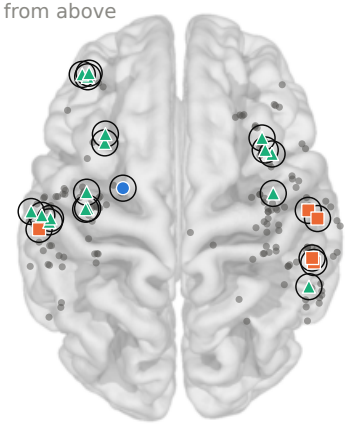
left hemisphere



right hemisphere



from above



- other electrodes of the cluster
- EL046
- EL048
- PAT_3975
- circled = position changed by the fix

Circled electrodes

patient	in the cluster	moved
EL046	1	1
EL048	5	5
PAT_3975	20	17

Where they are drawn

Desikan label of the nearest pial vertex

region	before	after
middle temporal	4	9
insula	1	7
superior frontal	4	–
medial wall (no label)	3	1
rostral middle frontal	–	3
entorhinal	2	–
fusiform	–	2
inferior temporal	2	–
precentral	2	–
6 more regions	5	1

The whole cluster, by lobe

electrodes

lobe	before	after
frontal	27	21
temporal	84	86
parietal	6	7
occipital	2	1
insula	14	20
cingulate	2	2
medial wall (no label)	29	27

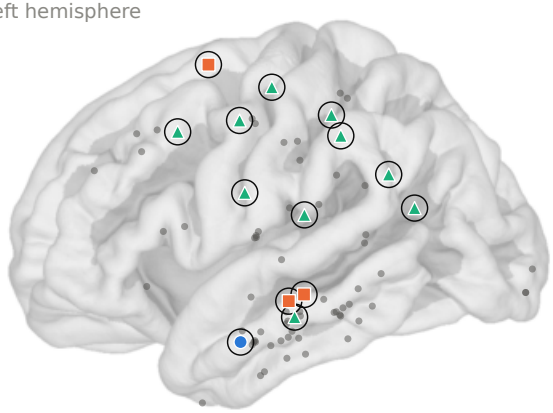
c3 sentence-stimulus increase

main solution: convex NMF, 15 bands, K = 7

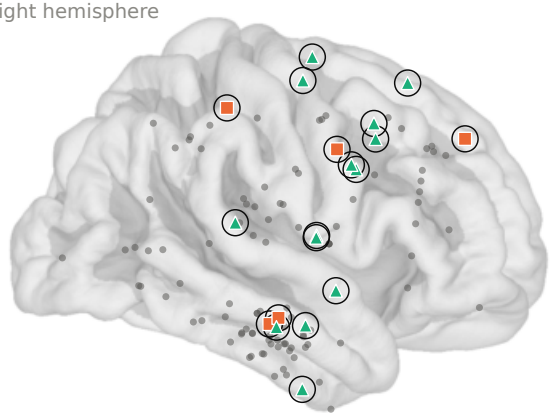
194 electrodes from 25 patients · 44 belong to the three patients · 33 were drawn at a wrong position (17 % of the cluster), median move 48 mm

BEFORE the cluster as it was drawn until now

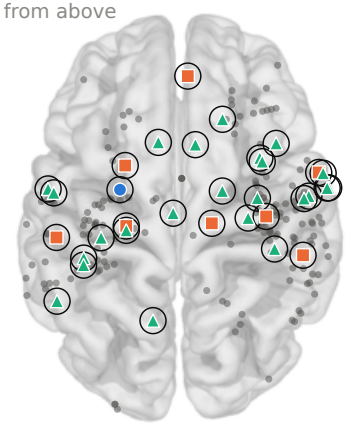
left hemisphere



right hemisphere

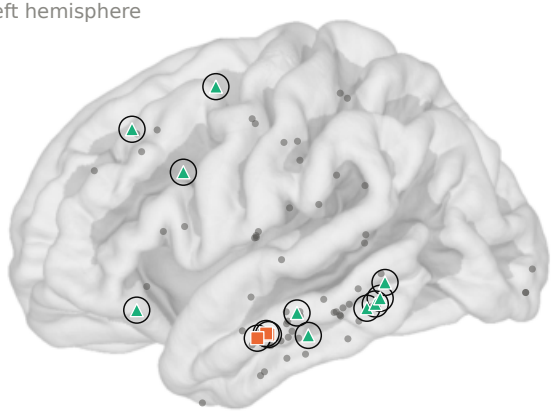


from above

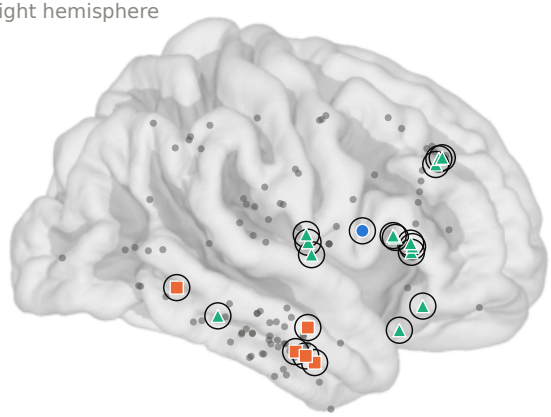


AFTER the cluster with the corrected positions

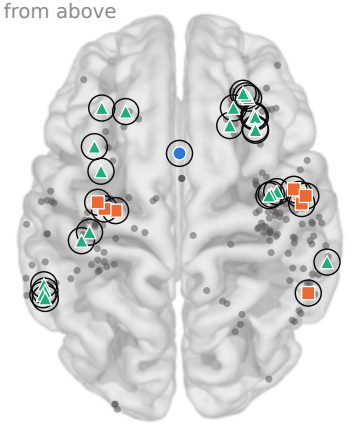
left hemisphere



right hemisphere



from above



• other electrodes of the cluster ◉ EL046 ◉ EL048 ◉ PAT_3975 circled = position changed by the fix

Circled electrodes

patient	in the cluster	moved
EL046	1	1
EL048	8	8
PAT_3975	35	24

Where they are drawn

Desikan label of the nearest pial vertex

region	before	after
middle temporal	2	7
superior frontal	6	–
insula	1	5
medial wall (no label)	5	5
lateral orbitofrontal	–	4
precentral	4	–
rostral middle frontal	–	4
caudal middle frontal	3	2
inferior temporal	1	3
8 more regions	11	3

The whole cluster, by lobe

electrodes

lobe	before	after
frontal	44	42
temporal	69	74
parietal	32	25
occipital	8	8
insula	9	13
cingulate	2	2
medial wall (no label)	30	30

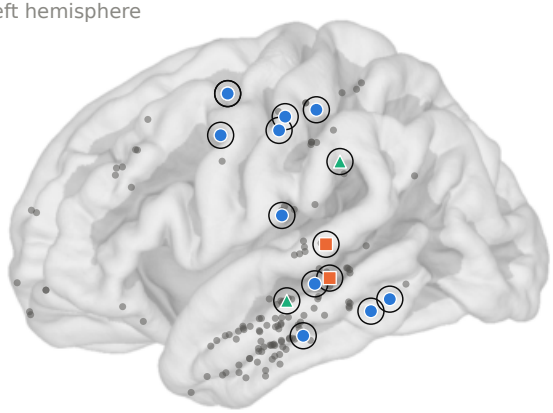
c4 low-frequency increase

main solution: convex NMF, 15 bands, K = 7

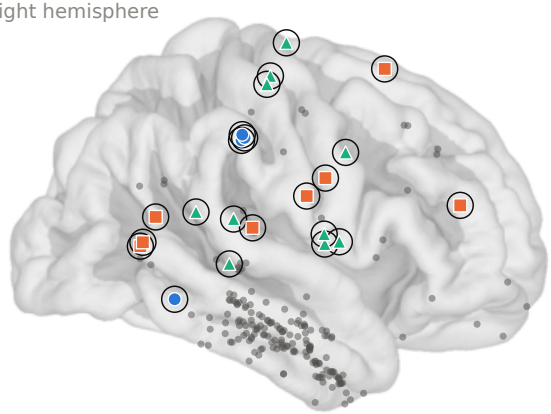
349 electrodes from 29 patients · 39 belong to the three patients · 37 were drawn at a wrong position (11 % of the cluster), median move 60 mm

BEFORE the cluster as it was drawn until now

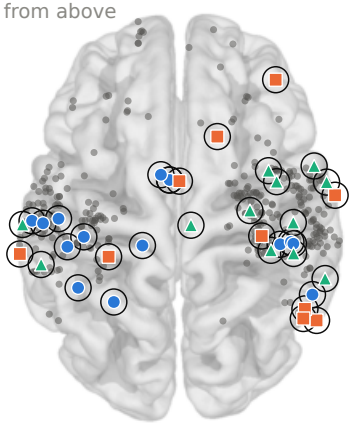
left hemisphere



right hemisphere

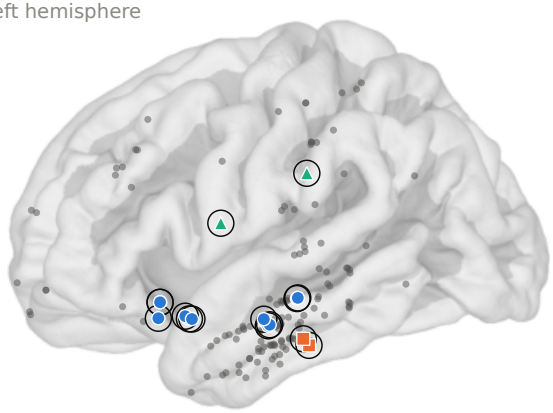


from above

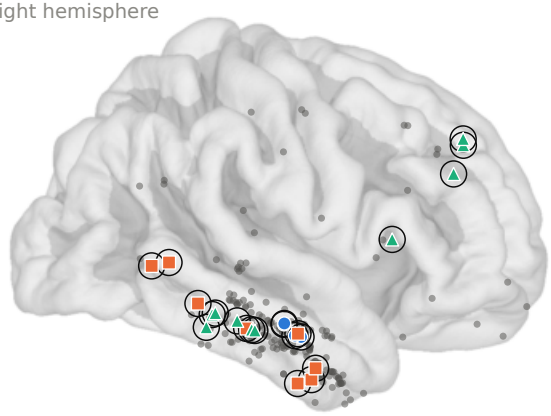


AFTER the cluster with the corrected positions

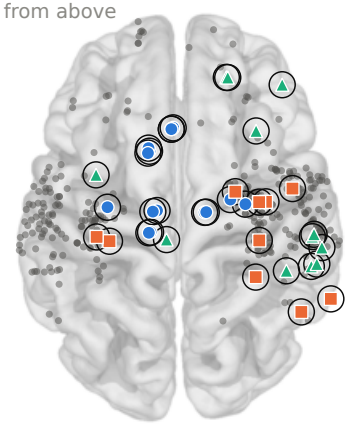
left hemisphere



right hemisphere



from above



Circled electrodes

patient	in the cluster	moved
EL046	15	15
EL048	10	10
PAT_3975	14	12

Where they are drawn

Desikan label of the nearest pial vertex

region	before	after
medial wall (no label)	3	9
middle temporal	3	7
postcentral	6	–
supramarginal	5	–
fusiform	3	3
insula	3	2
lateral orbitofrontal	–	3
medial orbitofrontal	–	3
precentral	3	–
9 more regions	11	10

The whole cluster, by lobe

electrodes

lobe	before	after
frontal	44	45
temporal	210	216
parietal	25	13
insula	11	10
cingulate	1	1
medial wall (no label)	58	64

• other electrodes of the cluster ◯ EL046 ◯ EL048 △ PAT_3975 circled = position changed by the fix

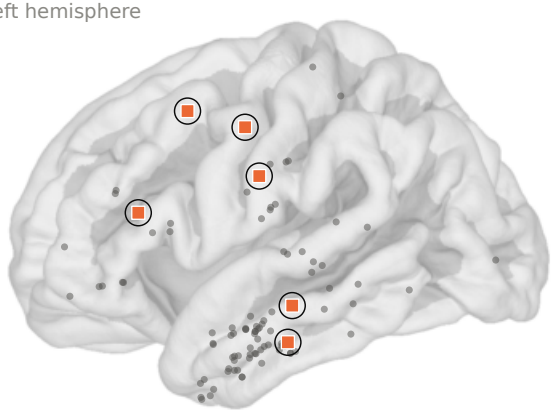
c5 response broadband increase

main solution: convex NMF, 15 bands, K = 7

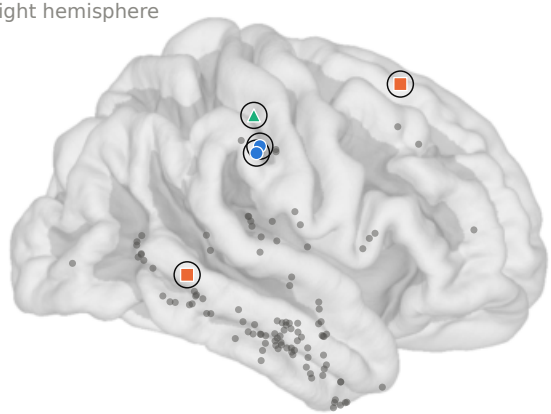
206 electrodes from 18 patients · 11 belong to the three patients · 11 were drawn at a wrong position (5 % of the cluster), median move 65 mm

BEFORE the cluster as it was drawn until now

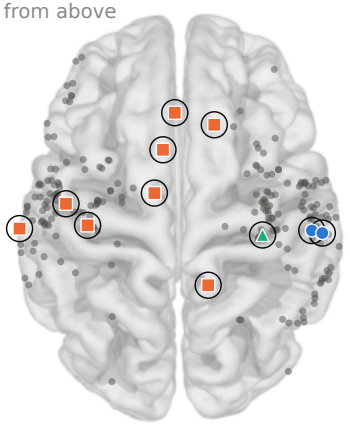
left hemisphere



right hemisphere

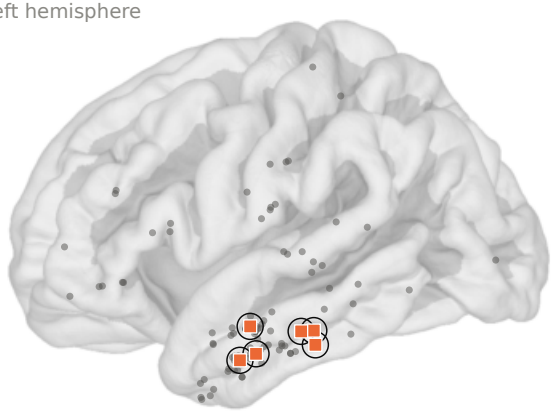


from above

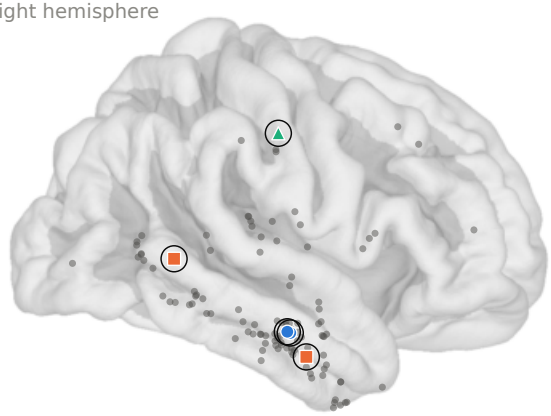


AFTER the cluster with the corrected positions

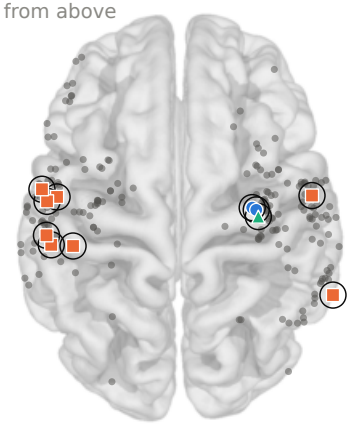
left hemisphere



right hemisphere



from above



- other electrodes of the cluster
- EL046
- EL048
- PAT_3975
- circled = position changed by the fix

Circled electrodes

patient	in the cluster	moved
EL046	2	2
EL048	8	8
PAT_3975	1	1

Where they are drawn

Desikan label of the nearest pial vertex

region	before	after
middle temporal	1	5
inferior temporal	–	3
superior frontal	3	–
medial wall (no label)	–	2
supramarginal	2	–
caudal ant. cingulate	1	–
fusiform	1	–
lingual	1	–
postcentral	1	–
1 more regions	1	1

The whole cluster, by lobe

electrodes

lobe	before	after
frontal	19	16
temporal	132	138
parietal	14	11
occipital	6	5
insula	16	16
cingulate	1	0
medial wall (no label)	18	20

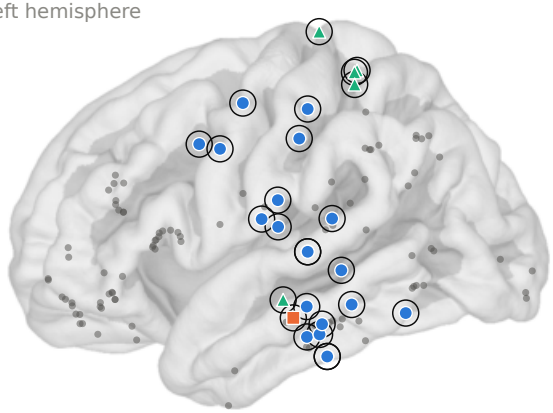
c6 response low-freq. decrease

main solution: convex NMF, 15 bands, K = 7

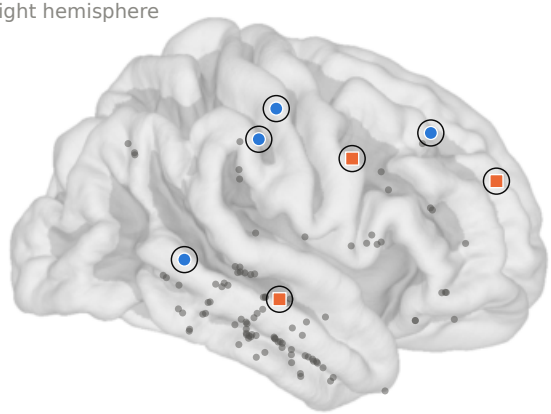
223 electrodes from 23 patients · 35 belong to the three patients · 33 were drawn at a wrong position (15 % of the cluster), median move 61 mm

BEFORE the cluster as it was drawn until now

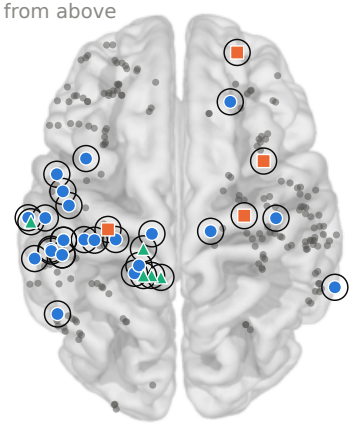
left hemisphere



right hemisphere

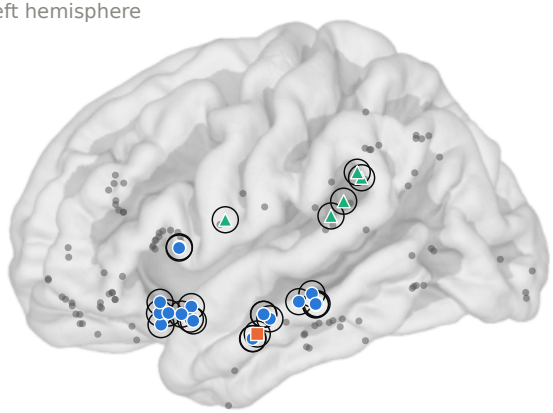


from above

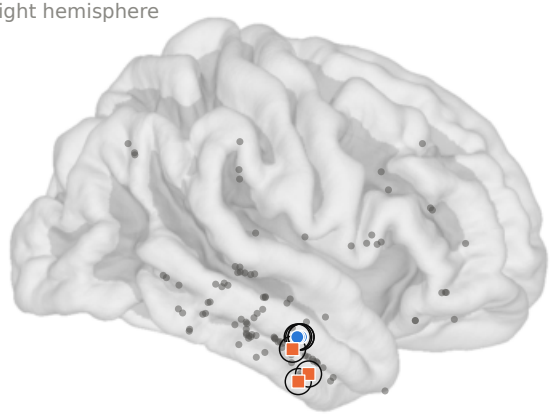


AFTER the cluster with the corrected positions

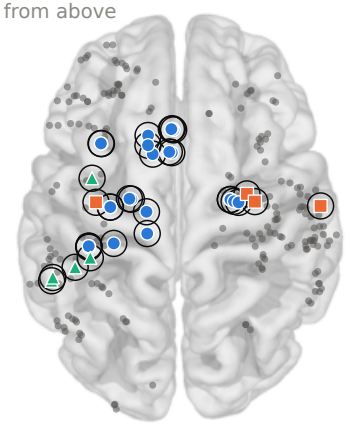
left hemisphere



right hemisphere



from above



• other electrodes of the cluster ◯ EL046 ◯ EL048 △ PAT_3975 circled = position changed by the fix

Circled electrodes

patient	in the cluster	moved
EL046	24	24
EL048	4	4
PAT_3975	7	5

Where they are drawn

Desikan label of the nearest pial vertex

region	before	after
medial wall (no label)	4	10
entorhinal	–	6
medial orbitofrontal	–	6
postcentral	6	–
inferior temporal	5	–
supramarginal	–	4
insula	–	3
paracentral	3	–
superior temporal	3	–
9 more regions	12	4

The whole cluster, by lobe

electrodes

lobe	before	after
frontal	65	64
temporal	96	92
parietal	25	23
occipital	7	7
insula	14	17
cingulate	2	0
medial wall (no label)	14	20

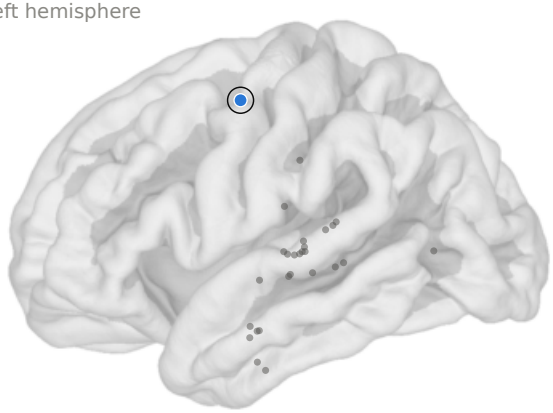
c0

convex NMF, 15 bands each z-scored, K = 7

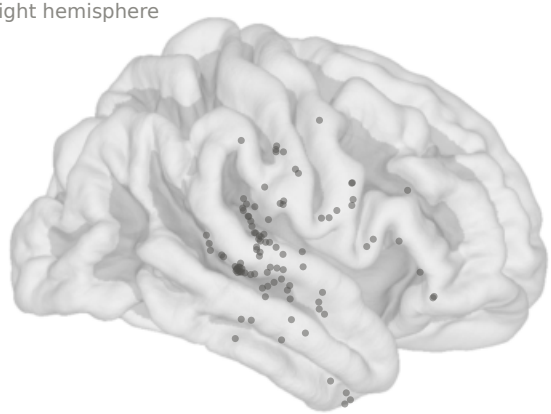
126 electrodes from 20 patients · 1 belong to the three patients · 1 were drawn at a wrong position (1 % of the cluster), median move 67 mm

BEFORE the cluster as it was drawn until now

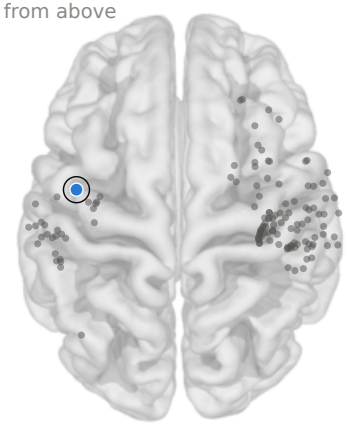
left hemisphere



right hemisphere

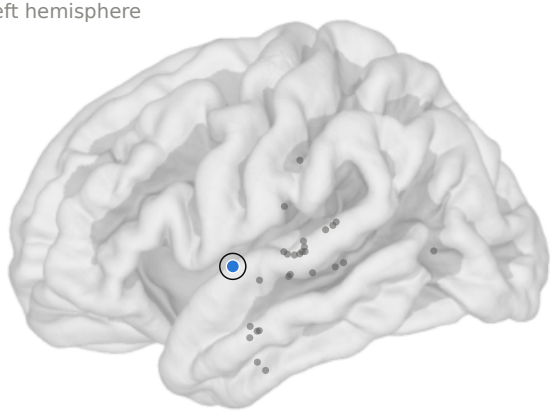


from above

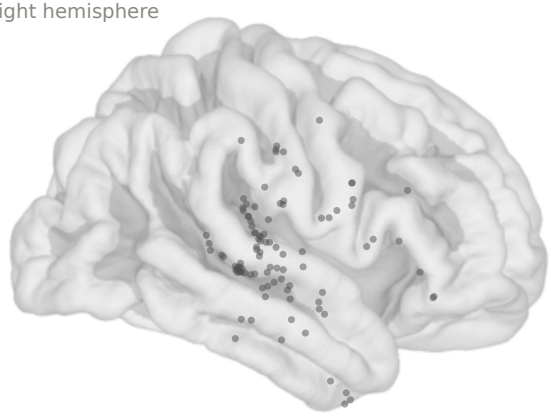


AFTER the cluster with the corrected positions

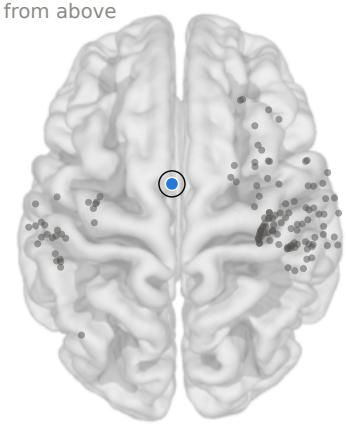
left hemisphere



right hemisphere



from above



Circled electrodes

patient	in the cluster	moved
EL046	1	1
EL048	0	0
PAT_3975	0	0

Where they are drawn

Desikan label of the nearest pial vertex

region	before	after
medial wall (no label)	–	1
precentral	1	–

The whole cluster, by lobe

electrodes

lobe	before	after
frontal	16	15
temporal	68	68
parietal	11	11
occipital	1	1
insula	24	24
medial wall (no label)	6	7

• other electrodes of the cluster ◉ EL046 ◉ EL048 ◉ PAT_3975 circled = position changed by the fix

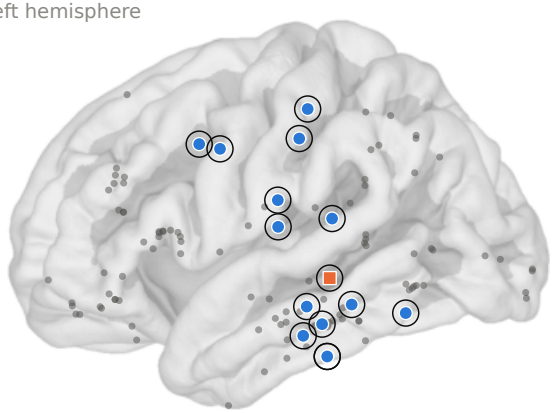
c1

convex NMF, 15 bands each z-scored, K = 7

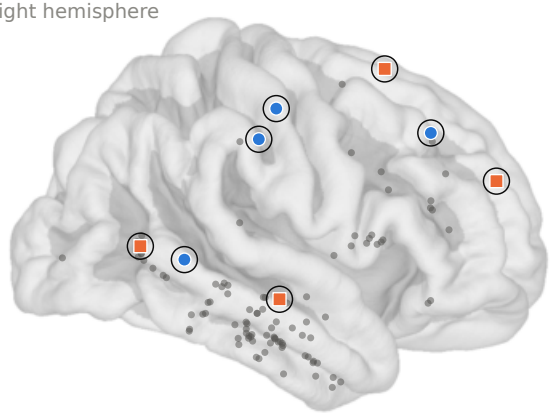
217 electrodes from 24 patients · 23 belong to the three patients · 23 were drawn at a wrong position (11 % of the cluster), median move 61 mm

BEFORE the cluster as it was drawn until now

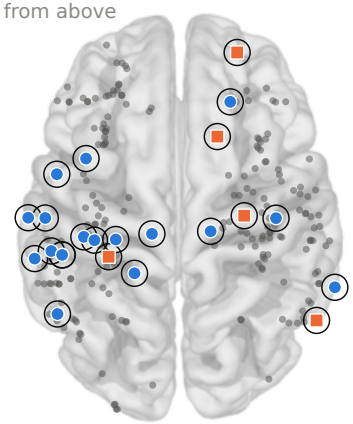
left hemisphere



right hemisphere

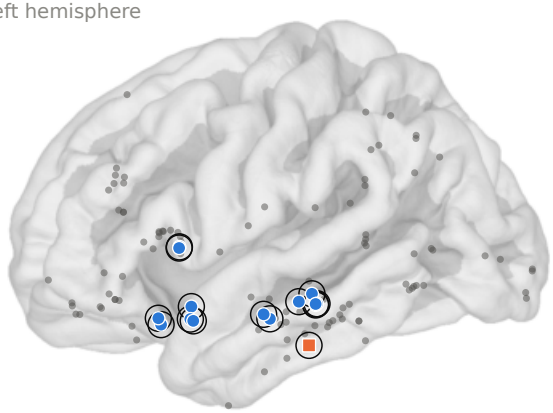


from above

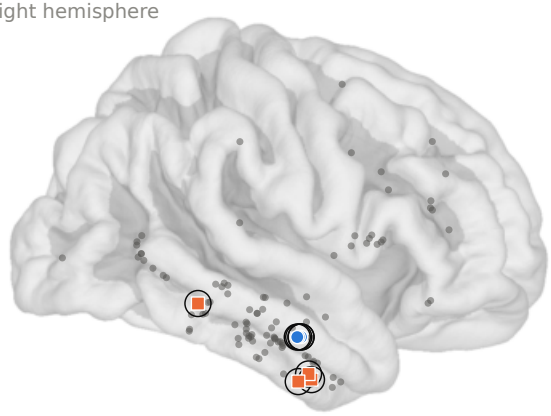


AFTER the cluster with the corrected positions

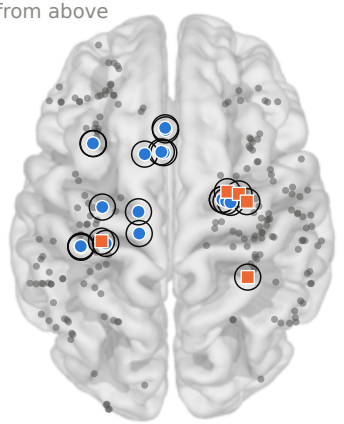
left hemisphere



right hemisphere



from above



• other electrodes of the cluster ◯ EL046 ◯ EL048 △ PAT_3975 circled = position changed by the fix

Circled electrodes

patient	in the cluster	moved
EL046	18	18
EL048	5	5
PAT_3975	0	0

Where they are drawn

Desikan label of the nearest pial vertex

region	before	after
medial wall (no label)	3	8
entorhinal	–	5
medial orbitofrontal	–	5
inferior temporal	4	–
postcentral	4	–
fusiform	1	2
insula	–	2
middle temporal	2	–
posterior cingulate	2	–
6 more regions	7	1

The whole cluster, by lobe

electrodes

lobe	before	after
frontal	50	50
temporal	92	91
parietal	15	11
occipital	13	13
insula	18	20
cingulate	2	0
medial wall (no label)	27	32

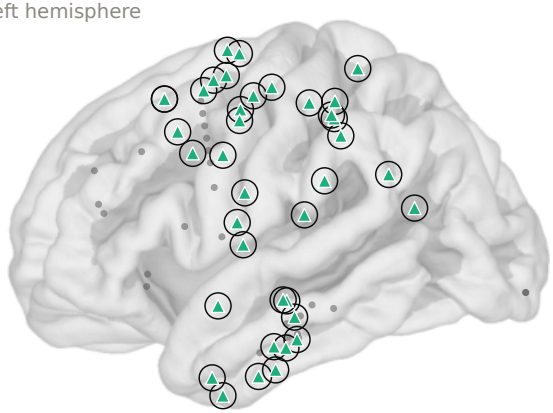
c2

convex NMF, 15 bands each z-scored, K = 7

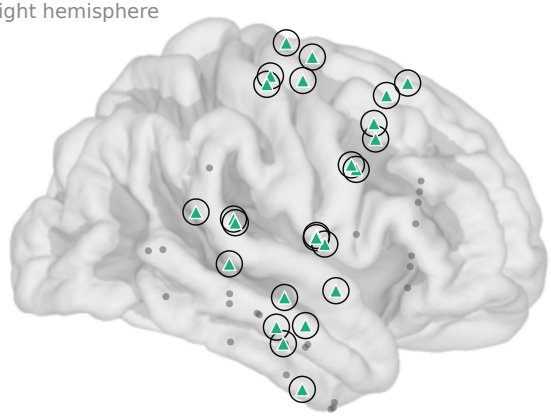
112 electrodes from 11 patients · 88 belong to the three patients · 62 were drawn at a wrong position (55 % of the cluster), median move 48 mm

BEFORE the cluster as it was drawn until now

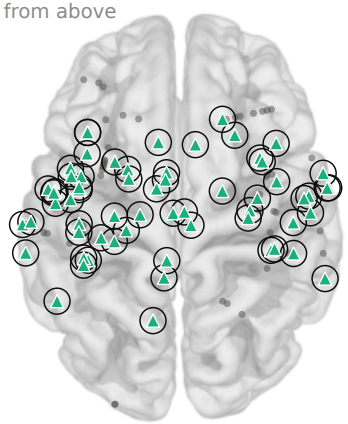
left hemisphere



right hemisphere

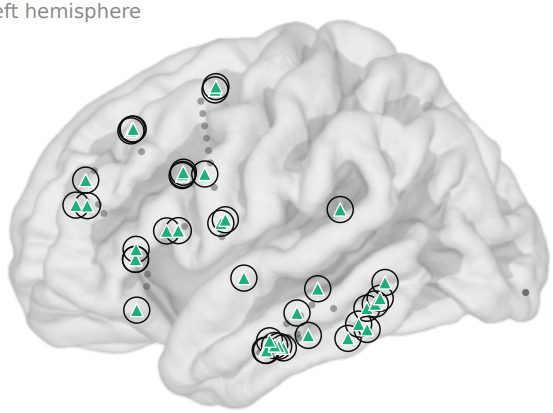


from above

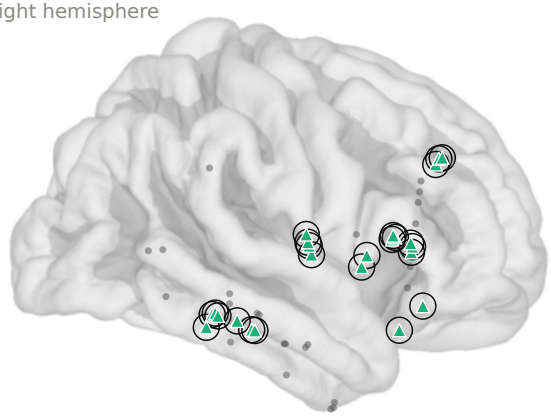


AFTER the cluster with the corrected positions

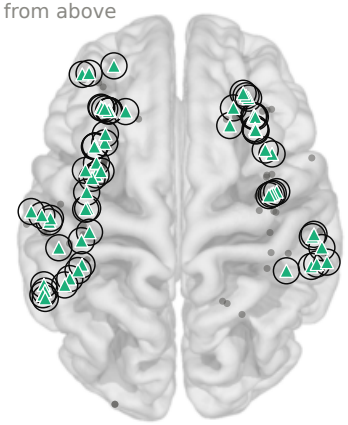
left hemisphere



right hemisphere



from above



• other electrodes of the cluster ◉ EL046 ◉ EL048 ◉ PAT_3975 circled = position changed by the fix

Circled electrodes

patient	in the cluster	moved
EL046	0	0
EL048	0	0
PAT_3975	88	62

Where they are drawn

Desikan label of the nearest pial vertex

region	before	after
insula	2	14
middle temporal	5	12
precentral	10	–
rostral middle frontal	–	9
superior frontal	9	–
caudal middle frontal	7	6
inferior temporal	5	6
lateral orbitofrontal	–	6
postcentral	5	–
9 more regions	19	9

The whole cluster, by lobe

electrodes

lobe	before	after
frontal	51	45
temporal	33	38
parietal	11	2
occipital	4	4
insula	4	16
cingulate	1	1
medial wall (no label)	8	6

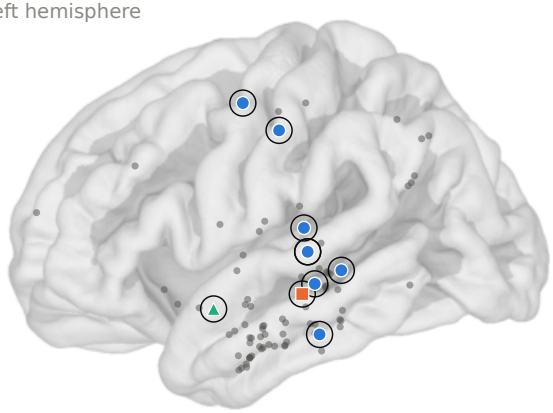
c3

convex NMF, 15 bands each z-scored, K = 7

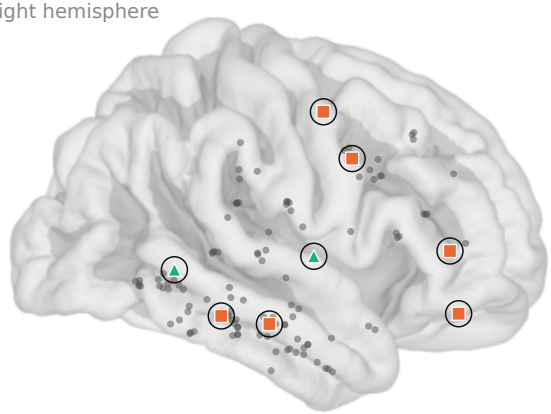
204 electrodes from 16 patients · 19 belong to the three patients · 18 were drawn at a wrong position (9 % of the cluster), median move 45 mm

BEFORE the cluster as it was drawn until now

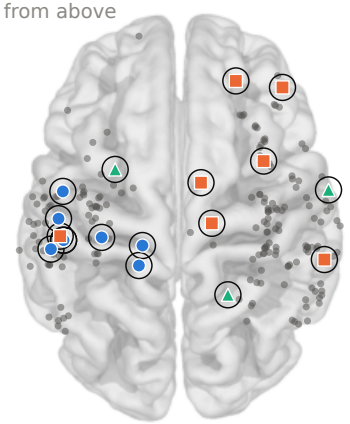
left hemisphere



right hemisphere

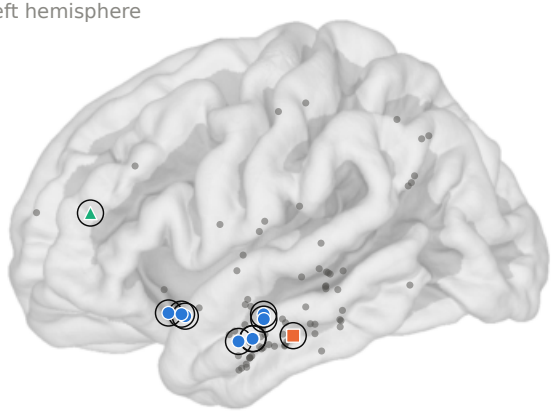


from above

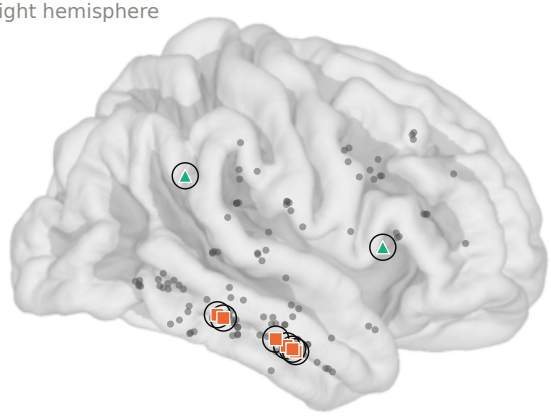


AFTER the cluster with the corrected positions

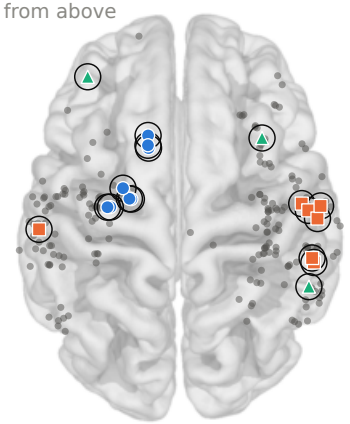
left hemisphere



right hemisphere



from above



- other electrodes of the cluster
- EL046
- EL048
- PAT_3975
- circled = position changed by the fix

Circled electrodes

patient	in the cluster	moved
EL046	8	8
EL048	7	7
PAT_3975	4	3

Where they are drawn

Desikan label of the nearest pial vertex

region	before	after
middle temporal	2	7
medial wall (no label)	4	3
lateral orbitofrontal	1	3
superior temporal	3	–
entorhinal	–	2
caudal middle frontal	1	–
inferior temporal	1	–
insula	1	1
lingual	1	–
6 more regions	4	2

The whole cluster, by lobe

electrodes

lobe	before	after
frontal	27	26
temporal	104	107
parietal	18	18
occipital	2	1
insula	19	19
cingulate	2	2
medial wall (no label)	32	31

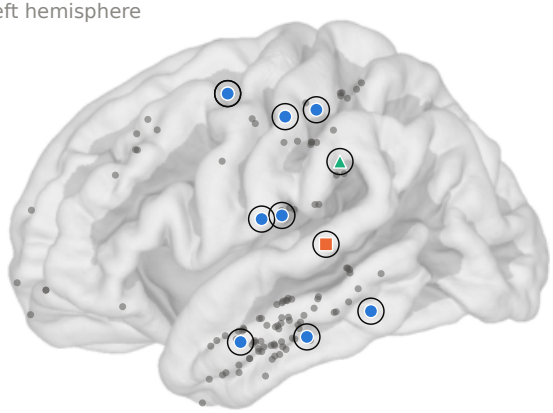
c4

convex NMF, 15 bands each z-scored, K = 7

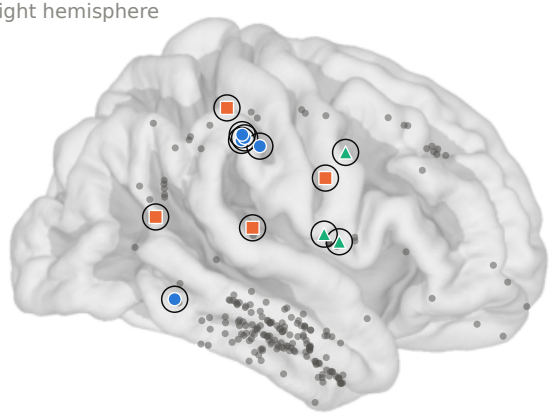
322 electrodes from 28 patients · 24 belong to the three patients · 23 were drawn at a wrong position (7 % of the cluster), median move 61 mm

BEFORE the cluster as it was drawn until now

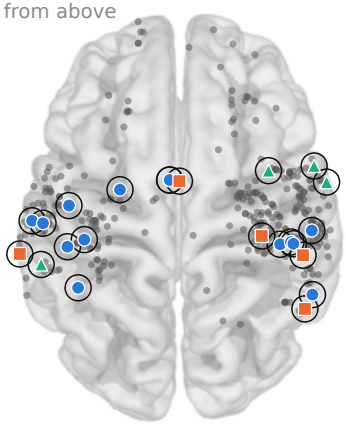
left hemisphere



right hemisphere

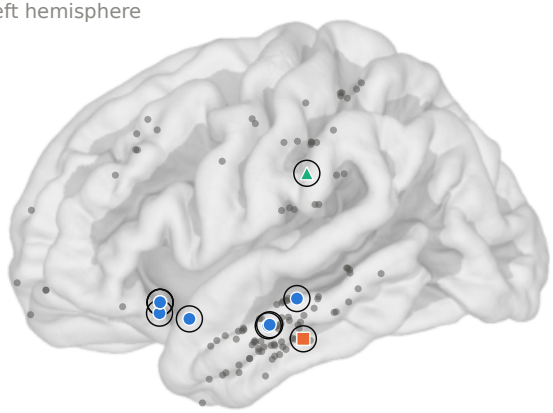


from above

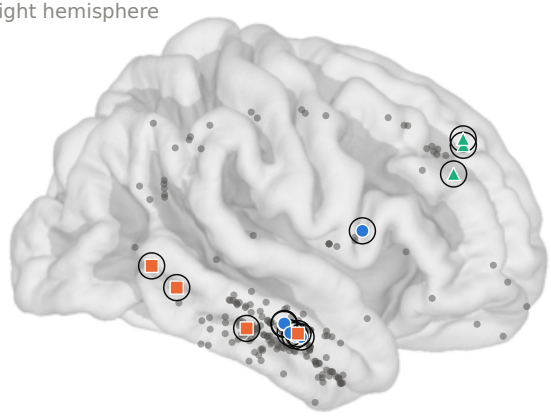


AFTER the cluster with the corrected positions

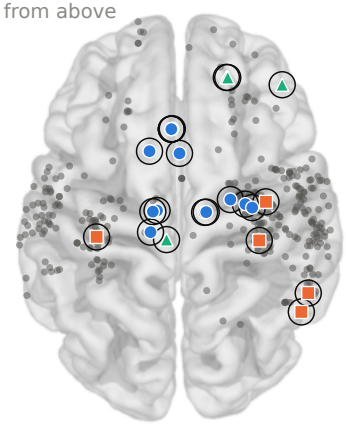
left hemisphere



right hemisphere



from above



• other electrodes of the cluster ◯ EL046 ◯ EL048 △ PAT_3975 circled = position changed by the fix

Circled electrodes

patient	in the cluster	moved
EL046	14	14
EL048	5	5
PAT_3975	5	4

Where they are drawn

Desikan label of the nearest pial vertex

region	before	after
medial wall (no label)	2	9
supramarginal	6	–
medial orbitofrontal	–	4
postcentral	4	–
fusiform	2	1
insula	2	–
middle temporal	–	2
precentral	2	–
superior frontal	2	2
8 more regions	3	5

The whole cluster, by lobe

electrodes

lobe	before	after
frontal	45	49
temporal	164	165
parietal	38	27
occipital	1	1
insula	9	7
cingulate	1	2
medial wall (no label)	64	71

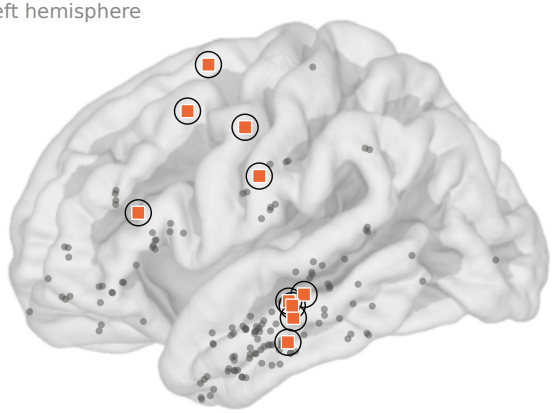
c5

convex NMF, 15 bands each z-scored, K = 7

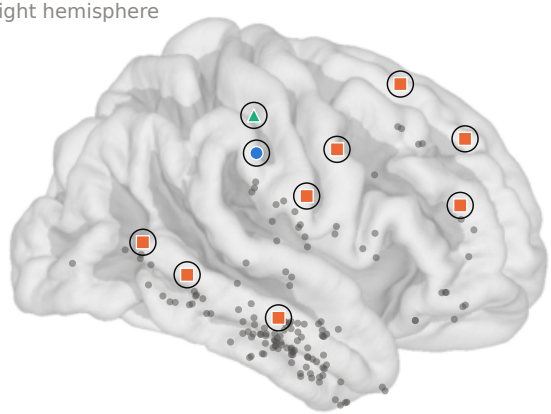
299 electrodes from 20 patients · 20 belong to the three patients · 20 were drawn at a wrong position (7 % of the cluster), median move 59 mm

BEFORE the cluster as it was drawn until now

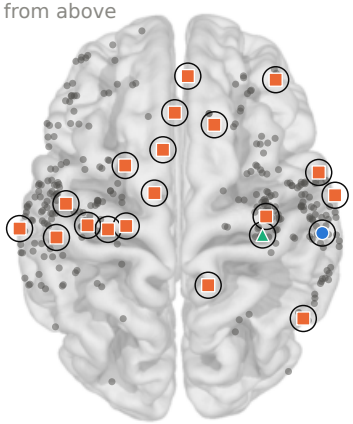
left hemisphere



right hemisphere

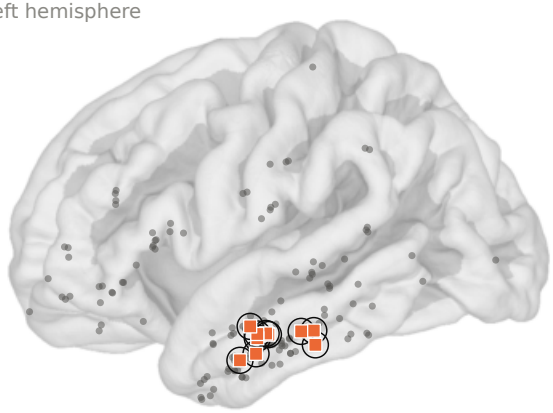


from above

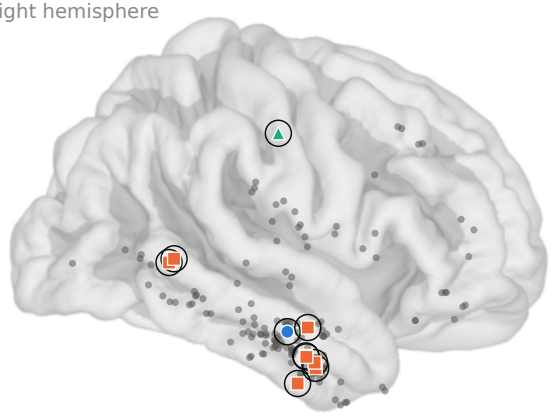


AFTER the cluster with the corrected positions

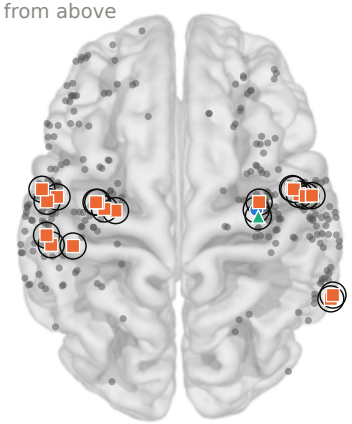
left hemisphere



right hemisphere



from above



- other electrodes of the cluster
- EL046
- EL048
- PAT_3975
- circled = position changed by the fix

Circled electrodes

patient	in the cluster	moved
EL046	1	1
EL048	18	18
PAT_3975	1	1

Where they are drawn

Desikan label of the nearest pial vertex

region	before	after
middle temporal	3	9
medial wall (no label)	3	5
superior frontal	5	–
inferior temporal	–	4
postcentral	2	–
precentral	2	1
caudal ant. cingulate	1	–
fusiform	1	1
lingual	1	–
2 more regions	2	0

The whole cluster, by lobe

electrodes

lobe	before	after
frontal	57	50
temporal	180	190
parietal	15	12
occipital	6	5
insula	18	18
cingulate	1	0
medial wall (no label)	22	24

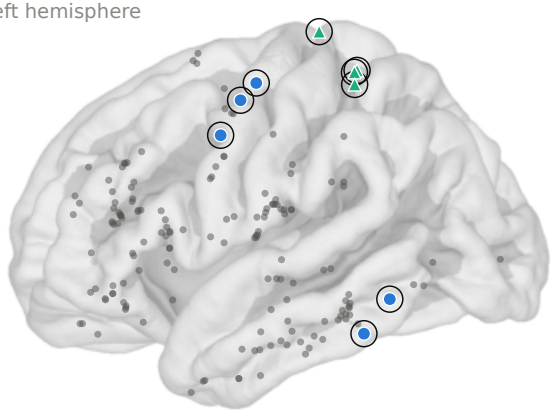
c6

convex NMF, 15 bands each z-scored, K = 7

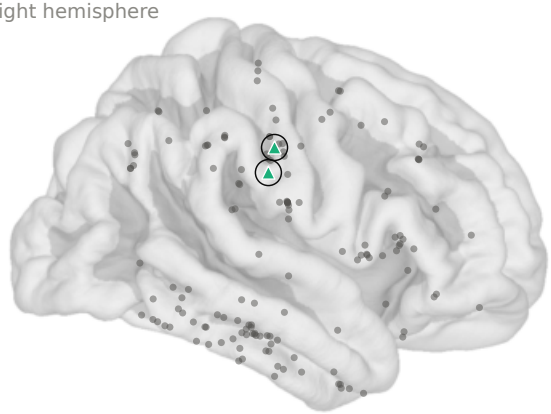
307 electrodes from 19 patients · 17 belong to the three patients · 11 were drawn at a wrong position (4 % of the cluster), median move 55 mm

BEFORE the cluster as it was drawn until now

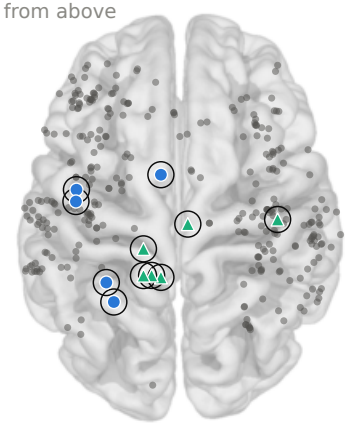
left hemisphere



right hemisphere

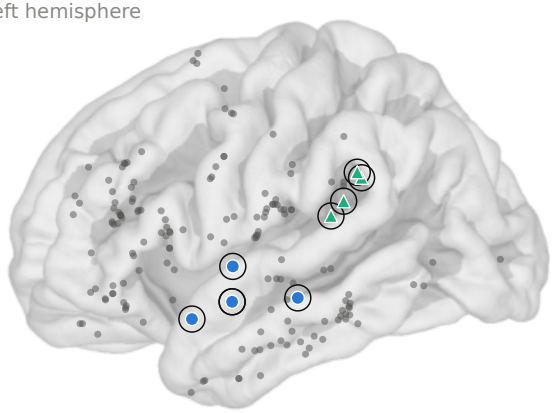


from above

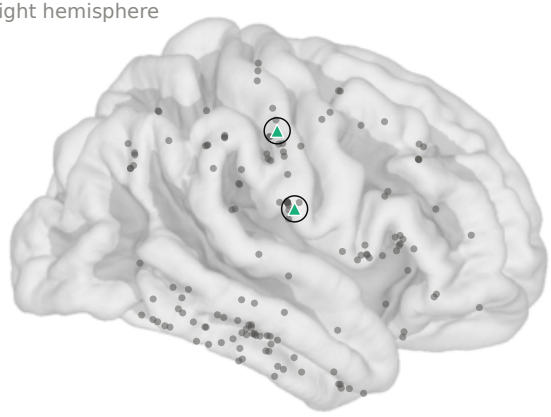


AFTER the cluster with the corrected positions

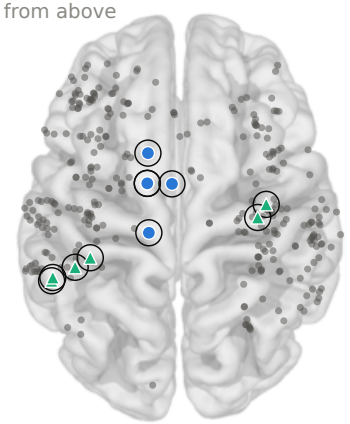
left hemisphere



right hemisphere



from above



Circled electrodes

patient	in the cluster	moved
EL046	5	5
EL048	0	0
PAT_3975	12	6

Where they are drawn

Desikan label of the nearest pial vertex

region	before	after
medial wall (no label)	–	4
supramarginal	–	4
paracentral	3	–
fusiform	2	–
postcentral	2	–
posterior cingulate	2	–
precentral	2	1
insula	–	1
lateral orbitofrontal	–	1

The whole cluster, by lobe

electrodes

lobe	before	after
frontal	107	104
temporal	97	95
parietal	53	55
occipital	3	3
insula	34	35
cingulate	8	6
medial wall (no label)	5	9

• other electrodes of the cluster ◉ EL046 ◉ EL048 ◉ PAT_3975 circled = position changed by the fix

Every run

The same 158 electrodes moved in every run; only the cluster they sit in differs. Each run at the K the cluster visualizer shows.

run	K	electrodes drawn at a wrong position / electrodes in the cluster	most affected cluster
Convex NMF, 2 bands	7	c0 15/185 c1 11/201 c2 21/264 c3 12/190 c4 39/264 c5 43/310 c6 17/173	c4 15 %
K-means, 2 bands	6	c0 17/433 c1 29/218 c2 30/169 c3 30/236 c4 41/343 c5 11/188	c2 18 %
Ward, 2 bands	7	c0 63/297 c1 5/144 c2 2/56 c3 11/91 c4 21/506 c5 24/254 c6 32/239	c0 21 %
Convex NMF, 5 bands	7	c0 12/188 c1 15/208 c2 28/305 c3 35/221 c4 18/252 c5 11/214 c6 39/199	c6 20 %
K-means, 5 bands	5	c0 30/249 c1 21/411 c2 23/314 c3 58/255 c4 26/358	c3 23 %
Ward, 5 bands	6	c0 27/424 c1 66/97 c2 17/187 c3 10/332 c4 16/132 c5 22/415	c1 68 %
Convex NMF, 15 bands	7	c0 18/316 c1 3/135 c2 23/164 c3 33/194 c4 37/349 c5 11/206 c6 33/223	c3 17 %
K-means, 15 bands	5	c0 25/363 c1 30/431 c2 44/219 c3 30/347 c4 29/227	c2 20 %
Ward, 15 bands	25	158 moved over 25 clusters	c12 70 %
Convex NMF, HFA only	7	c0 40/168 c1 22/256 c2 4/81 c3 14/233 c4 7/225 c5 17/166 c6 54/458	c0 24 %
K-means, HFA only	5	c0 57/381 c1 15/162 c2 45/600 c3 19/178 c4 22/266	c0 15 %
Ward, HFA only	8	c0 0/59 c1 18/90 c2 0/68 c3 23/155 c4 23/130 c5 61/353 c6 9/224 c7 24/508	c1 20 %
Convex NMF, 15 bands, each z-scored	7	c0 1/126 c1 23/217 c2 62/112 c3 18/204 c4 23/322 c5 20/299 c6 11/307	c2 55 %
K-means, 15 bands, each z-scored	5	c0 25/332 c1 21/395 c2 12/296 c3 68/273 c4 32/291	c3 25 %
Convex NMF, 15 bands, per-band shape	7	c0 21/187 c1 7/292 c2 25/217 c3 2/157 c4 66/153 c5 8/318 c6 29/263	c4 43 %
K-means, 15 bands, per-band shape	9	c0 32/173 c1 2/192 c2 17/163 c3 9/302 c4 5/280 c5 4/170 c6 22/134 c7 67/102 c8 0/71	c7 66 %